Auditing an Interpretable Weather–Symptom–Disease Cascade: Leakage-Resistant Evaluation and Error Propagation

Yun Tao, Sijie Liu, Junhan Yi, Yumeng Jin, Yinuo Pang
Artificial Intelligence Programme
Beijing Normal–Hong Kong Baptist University
Zhuhai, Guangdong, China

***Abstract*—Semantic bottlenecks can make machine-learning pipelines easier to inspect, but their end-to-end value depends on whether the intermediate concepts can be predicted reliably. We audit a two-stage weather–symptom–disease cascade on the public Weather-related Disease Prediction Dataset. The released file contains 5,200 rows, 219 of which are exact duplicates; one of 37 symptom columns is constant. Our primary analysis therefore uses 4,981 unique rows and 36 estimable symptoms. We construct a locked, weather-signature-disjoint test set (3,984 development and 997 test rows), generate five-fold group-disjoint out-of-fold symptom predictions for disease-stage training, and select all thresholds without test-label access. On the locked test set, weather-only XGBoost attains a symptom micro-F1 of 0.1889. The best direct weather-only disease model reaches 20.16% accuracy (95% bootstrap confidence interval [17.75%, 22.67%]), close to the 19.36% class-prior baseline, whereas the prespecified XGBoost-to-TabNet cascade reaches 11.94% [10.03%, 13.94%]. Adding age, gender, and documented pre-existing conditions raises direct XGBoost accuracy to 41.12% [38.01%, 44.03%]; a context-aware XGBoost symptom bottleneck followed by XGBoost reaches 39.62% [36.61%, 42.73%]. In contrast, TabNet supplied with true symptoms reaches 96.29% [95.09%, 97.39%]. The resulting 84.35-percentage-point Oracle-to-deployable gap identifies intermediate prediction error as the dominant limitation. The cascade is thus a useful model-audit case study, but the present data do not support clinical, causal, or early-warning claims. External validation with patient identifiers, timestamps, locations, and independent outcomes is required.**

***Index Terms*—clinical prediction model audit, semantic bottleneck, error propagation, weather and health, multilabel classification, calibration, reproducibility**

I. Introduction

Weather and climate exposures are associated with substantial health burden, including heat-related mortality and changes in infectious and respiratory risk [1]. Machine learning can represent nonlinear relations among environmental, demographic, and health variables, but apparent predictive performance may be misleading when duplicate records, non-independent splits, in-sample intermediate predictions, or weak baselines are not handled explicitly.

A semantic bottleneck offers an intuitively attractive design. Instead of mapping weather directly to a disease label, a first model predicts named symptoms, and a second model maps the symptom representation to disease. Humans can inspect the intermediate concepts, and the two stages can be diagnosed separately. This interpretability argument, however, is not itself a performance guarantee. If weather carries little information about individual symptoms, the second stage receives a noisy representation and may underperform a direct classifier.

This study revises and audits such a two-stage cascade. The audit was motivated by three questions:

1) Can weather variables predict the released symptom labels under a group-disjoint, leakage-resistant protocol?
2) Does the predicted-symptom bottleneck improve disease classification over direct and class-prior baselines?
3) How much of the end-to-end error is attributable to the symptom prediction layer rather than the disease classifier?

Our contributions are methodological rather than clinical. We provide a checksum-verified data pipeline, remove exact duplicates, make units explicit, train the disease stage on out-of-fold predictions, lock the final test set, report uncertainty and calibration, and compare predicted, thresholded, true, and context-enriched symptom representations. The audit finds a large gap between true and predicted symptoms. This negative result is scientifically important: semantic interpretability is meaningful only when the intermediate concepts are observable or predictable at the intended decision time.

II. Related Work and Reporting Standards

A. Weather, health, and prediction

Attribution research has established that human-induced climate change has already increased heat-related mortality [1]. Such population-level associations motivate environmental health forecasting, but they do not imply that a small set of contemporaneous weather measurements can diagnose an individual disease. Prediction tasks must distinguish population risk, surveillance, and individual diagnosis, because each requires different outcomes, sampling frames, and validation designs.

B. Semantic bottlenecks and tabular models

Concept bottleneck models explicitly predict human-defined concepts before a final task, enabling intervention and inspection at the concept layer [2]. Our symptom layer has this structure, although the public data do not provide concept annotations with timing or clinical adjudication. XGBoost is a regularized gradient-boosting implementation effective for

TABLE I
DATA AUDIT AND PRIMARY SPLIT

Item	Count/value
Released rows / columns	5,200 / 51
Exact duplicate rows	219
Primary analysis rows	4,981
Development / locked test rows	3,984 / 997
Disease classes	11
Source / modeled symptom columns	37 / 36
Unique raw-weather signatures	3,588
Weather-signature overlap across split	0
Test rows sharing a symptom signature with development	960

heterogeneous tabular predictors [3]. TabNet uses sequential attention to select tabular features and provides instance-level feature masks [4]. We treat these algorithms as model families to be evaluated, not as evidence of clinical validity.

C. Transparent prediction-model evaluation

TRIPOD+AI emphasizes transparent reporting of data sources, participants, predictors, outcomes, analysis, performance, and limitations for prediction models using regression or machine learning [5]. PROBAST+AI similarly structures risk-of-bias and applicability assessment [6]. Following these principles, we report duplicate handling, units, split independence, missing metadata, baselines, confidence intervals, calibration, and the precise boundary between internal benchmarking and external validation.

III. DATA AND AUDIT FINDINGS

A. Source and provenance

We use version 1 of the Weather-related Disease Prediction Dataset, published on Zenodo under CC BY 4.0 [7]. The archived CSV is preserved unchanged and verified before each publication run with the official MD5 checksum: `4aa51b2bb76b45b2000ce71517a0fd1e`.

The file contains age, binary gender, temperature in degrees Celsius, humidity, wind speed in km/h, binary symptom and pre-existing-condition indicators, and an 11-class `prognosis` label. The diseases are arthritis, common cold, dengue, eczema, heart attack, heat stroke, influenza, malaria, migraine, sinusitis, and stroke.

B. Data-quality decisions

The source has no missing cells, but 219 rows are exact duplicates. The primary analysis removes them before splitting. Two columns represent pain behind the eyes with near-duplicate names; they are merged by row-wise maximum into one concept. The `shivering` column is zero in all 5,200 rows, so its positive-class performance is mathematically unidentifiable. It remains in the data-quality inventory but is excluded from model fitting and macro averaging, leaving 36 estimable symptoms.

Humidity values range from approximately 0.37 to 1.00. Although the archive description calls humidity a percentage, we interpret the stored values as fractions and convert them to percent only inside named meteorological formulae. This decision is encoded once in the shared preprocessing function.

The source lacks patient identifiers, timestamps, locations, weather-station identifiers, institutions, and cohort indicators. We therefore cannot verify participant independence, temporal ordering, station matching, or geographic separation. Table I also shows that 960 of 997 test rows have a symptom signature observed in development. The weather split prevents exact weather-pattern leakage, but it cannot establish independent clinical generalization. As a split-sensitivity diagnostic, a naive stratified row split of the same size would share 280 raw-weather signatures between development and test, covering 377 of 997 test rows; the primary grouped split reduces both quantities to zero.

IV. METHODS

A. Weather representations

The raw weather representation is $\mathbf{x}_w = (T, \mathrm{RH}, v)$, where T is temperature in °C, RH is relative humidity, and v is wind speed in km/h. A full weather representation appends four deterministic indices:

- NOAA Rothfusz heat index, evaluated in Fahrenheit and converted back to Celsius in its warm–humid domain;
- humidex, using a Magnus dew-point approximation;
- Australian apparent temperature, converting wind to m/s; and
- temperature–humidity index

$$\mathrm{THI} = T - (0.55 - 0.0055\,\mathrm{RH})(T - 14.5). \qquad (1)$$

All transformations are deterministic. Scaling for logistic regression and MLP models is fitted within the corresponding training fold only.

B. Split, out-of-fold training, and thresholds

We hash the three raw weather values to define a weather signature. A seeded five-fold stratified group split contributes one fold as a locked test set and the other four as development data. Identical raw-weather signatures therefore cannot occur on both sides of the primary split. Within development, a second five-fold stratified group split produces out-of-fold (OOF) symptom probabilities.

For each symptom and Stage-1 model, the classification threshold is selected using development OOF predictions only. The deployable disease-stage models are fitted to OOF predicted symptom probabilities or OOF thresholded symptoms, never to in-sample Stage-1 outputs. The locked test labels enter only the final metric and statistical functions.

C. Models and comparisons

Stage 1 compares multilabel logistic regression (LR), a multilayer perceptron (MLP), and XGBoost. Direct disease baselines include an empirical class-prior classifier, multinomial LR, random forest (RF), MLP, and XGBoost. TabNet is the prespecified primary Stage-2 model; LR, RF, MLP, and XGBoost are architecture controls.

The experiments answer distinct questions:

TABLE II
LOCKED-TEST STAGE-1 SYMPTOM PERFORMANCE

Model	Inputs	Micro-F1	Macro-F1	Micro-AP	Hamming
LR	Weather + indices	0.1644	0.1495	0.1041	0.3843
MLP	Weather + indices	0.1575	0.1225	0.0914	0.5356
XGBoost	Raw weather	**0.1939**	0.1329	0.1616	0.2972
XGBoost	Weather + indices	0.1889	0.1323	0.1587	0.3073
XGBoost	Full pre-event context	**0.2429**	**0.2436**	**0.3754**	**0.2730**

1) **Weather-only cascade:** weather → predicted symptoms → disease.
2) **Binarization ablation:** symptom probabilities versus development-thresholded symptom indicators.
3) **Direct baselines:** weather → disease.
4) **Context extension:** age, gender, weather, and seven documented pre-existing conditions known before the modeled event.
5) **Skip connection:** context concatenated with predicted symptom probabilities before disease classification.
6) **Oracle upper bound:** true symptoms → TabNet.

The context extension changes the scientific question: it evaluates pre-event patient context plus weather, not weather alone. The Oracle is not deployable at a pre-symptom decision time and is used only to localize the bottleneck.

D. Metrics and statistical analysis

Stage-1 metrics are micro- and macro-F1, micro- and macro-average precision (AP), Hamming loss, and per-symptom precision, recall, F1, AP, prevalence, and threshold. Disease metrics are accuracy, balanced accuracy, macro- and weighted-F1, one-versus-rest macro AUROC and AP, negative log-likelihood, multiclass Brier score, 10-bin expected calibration error (ECE), per-class metrics, and confusion matrices.

We compute percentile 95% confidence intervals for accuracy and macro-F1 with 2,000 seeded nonparametric bootstrap replicates [8]. Exact McNemar tests compare paired locked-test correctness vectors [9]. Calibration is reported alongside discrimination because low ECE alone can occur in an uninformative model that stays close to class prevalence [10].

V. RESULTS

A. Stage-1 symptom prediction

Table II reports locked-test symptom prediction. XGBoost on the three raw weather variables has the highest weather-only micro-F1 (0.1939). Appending deterministic weather indices does not improve XGBoost (micro-F1 0.1889), suggesting that these transformations add little new information. LR has the highest weather-only macro-F1 (0.1495) but lower micro-AP. With pre-event context, XGBoost improves to micro-F1 0.2429 and micro-AP 0.3754. Performance remains modest, so the semantic bottleneck is noisy before disease-stage training begins.

B. End-to-end disease classification

Table III and Fig. 2 summarize the primary endpoint. The empirical class-prior classifier reaches 19.36% accuracy.

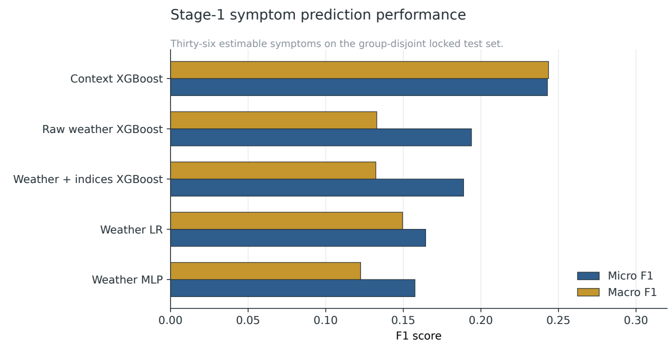

Fig. 1. Locked-test symptom prediction. The context extension is shown separately because it changes the available information.

The best weather-only direct model is MLP at 20.16%, with an interval overlapping the prior baseline. The prespecified weather-only XGBoost-to-TabNet cascade reaches 11.94% and macro-F1 0.0470. It does not improve on the direct baseline.

The direct context model is numerically strongest among deployable systems (41.12%). A context-aware XGBoost-to-XGBoost bottleneck retains 39.62%, and a skip connection reaches 40.22%. These small gaps show that a semantic representation can retain much of the context model's endpoint accuracy when both stages are matched to the tabular data, but it does not confer superiority. The prespecified TabNet second stage is particularly sensitive to noisy predicted symptom probabilities: replacing it with XGBoost raises the context bottleneck from 25.78% to 39.62%.

For the weather-only primary cascade, direct XGBoost is correct on 154 test rows where the cascade is wrong, whereas the cascade is correct on 93 rows where direct XGBoost is wrong (exact McNemar $p = 1.25 \times 10^{-4}$). Probability and thresholded-symptom TabNet variants have identical accuracy and symmetric discordance (98 versus 98; $p = 1.0$), so the data do not establish superiority of either representation. A non-significant test is not interpreted as equivalence.

C. Error propagation

The Oracle TabNet reaches 96.29% accuracy and macro-F1 0.9666. The same disease-stage family supplied with weather-predicted XGBoost symptom probabilities reaches only 11.94%. The 84.35-percentage-point gap (Fig. 3) demonstrates that the disease label is highly encoded in the recorded symptom pattern, but that weather alone cannot reconstruct the needed pattern in this evaluation.

D. Calibration, confusion, and attribution

Figure 4 plots confidence against observed accuracy. The direct context XGBoost has ECE 0.0416; the context bottleneck with XGBoost has ECE 0.0562. The weak weather-only TabNet cascade has ECE 0.0222, illustrating why calibration error must not be read without accuracy and classwise performance. The Oracle ECE is 0.1433 despite high accuracy, indicating that its probability scale would still require calibration for any decision use.

TABLE III
LOCKED-TEST DISEASE CLASSIFICATION ($n = 997$)

System	Representation	Accuracy	95% CI	Macro-F1
Class-prior baseline	Weather ignored	19.36%	[16.95, 21.77]	0.0295
Direct MLP	Weather + indices	20.16%	[17.75, 22.67]	0.0946
XGBoost → TabNet	Predicted symptoms, weather only	11.94%	[10.03, 13.94]	0.0470
Context XGBoost → TabNet	Context-predicted symptoms	25.78%	[22.97, 28.49]	0.1086
Context XGBoost → XGBoost	Context-predicted symptoms	39.62%	[36.61, 42.73]	0.2677
Skip connection XGBoost	Context + predicted symptoms	40.22%	[37.21, 43.33]	0.2744
Direct context XGBoost	Full pre-event context	**41.12%**	[38.01, 44.03]	**0.2865**
Oracle TabNet	True symptoms	96.29%	[95.09, 97.39]	0.9666

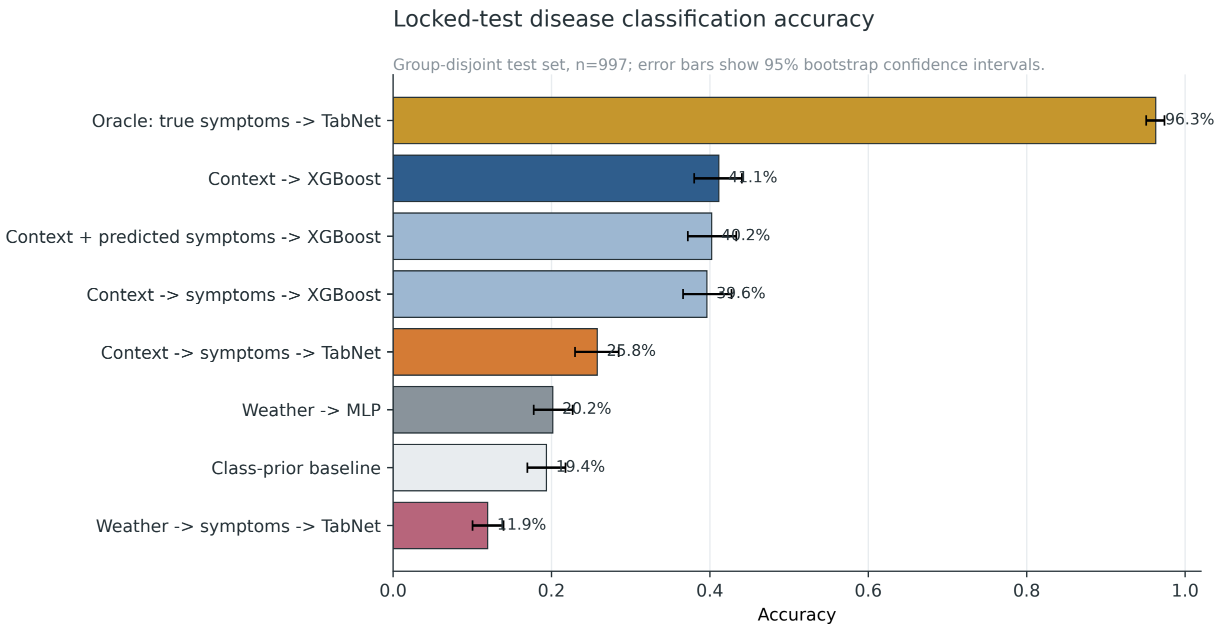

Fig. 2. Locked-test accuracy with 2,000-replicate bootstrap confidence intervals. Weather-only and context-extended experiments answer different scientific questions; the Oracle is an upper bound, not a deployable model.

The context symptom-bottleneck confusion matrix in Fig. 5 shows that a 39.62% aggregate accuracy does not translate to uniformly strong classwise recognition. This is reflected by the lower macro-F1 of 0.2677.

Permutation attribution is computed on held-out development-fold predictions, not on the locked test set. For direct context XGBoost, high blood pressure, HIV/AIDS, and nasal polyps have the largest macro-F1 decreases under permutation. In the Stage-1 context model, high blood pressure and HIV/AIDS dominate micro-AP importance. Aggregated TabNet masks in the Oracle model emphasize headache, chest pain, high fever, pain behind the eyes, vomiting, severe headache, rapid heart rate, and diarrhea (Fig. 6). These are associational model attributions, not causal effects or clinical risk factors.

VI. DISCUSSION

A. What the audit changes

The leakage-resistant results do not support the claim that three weather measurements are sufficient for accurate individual disease prediction. The best direct weather model is statistically compatible with a simple class-prior baseline, and the prespecified deployable cascade performs below that baseline. The more defensible contribution is an empirical account of semantic-bottleneck error propagation.

The Oracle result is useful but easy to misuse. It shows that the recorded symptoms nearly determine the released label under this split; it does not show that those symptoms are available before diagnosis, independently measured, or transferable to real clinical settings. Because 960 of 997 test rows share a symptom signature with development, the Oracle may partly learn a repeated label dictionary. It is an upper bound for this table, not a validated diagnostic system.

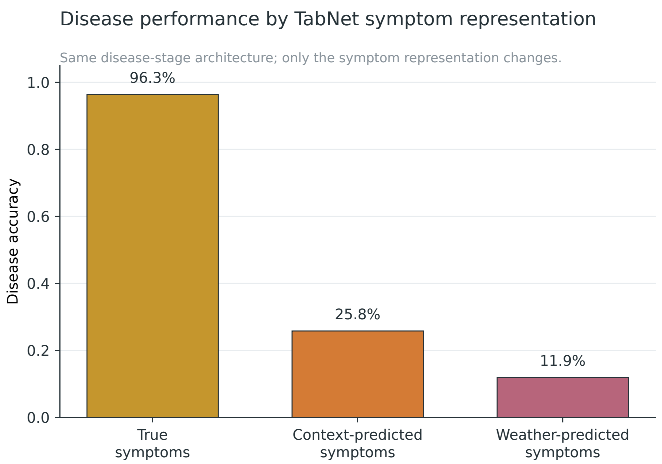

Fig. 3. Error propagation when the same TabNet disease-stage architecture is given weather-predicted, context-predicted, or true symptom representations.

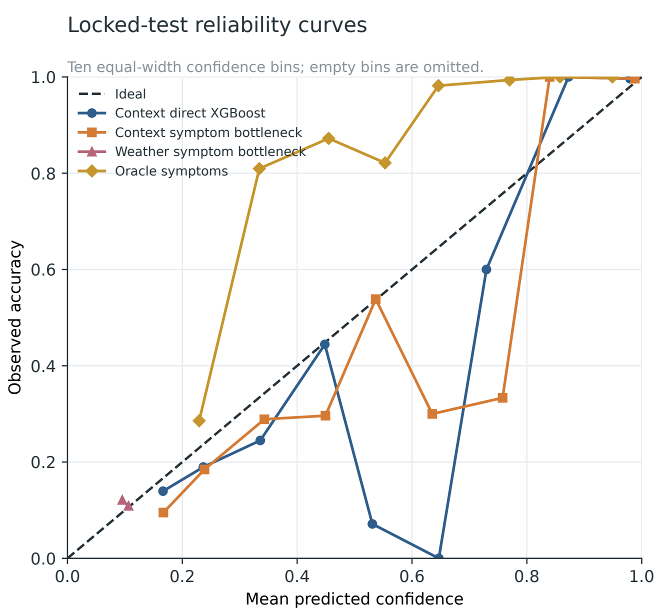

Fig. 4. Locked-test reliability curves. Marker area is proportional to bin count; ECE is reported with, not instead of, discrimination metrics.

B. Interpretability–performance trade-off

The symptom bottleneck makes the pipeline auditable: Stage-1 failure can be separated from disease-stage failure, thresholds can be inspected, and the effect of true versus predicted concepts can be quantified. With full pre-event context and XGBoost at both stages, it retains most of the direct model's endpoint accuracy. However, the bottleneck neither creates information nor guarantees robustness. The skip connection recovers little beyond direct context, and TabNet underperforms XGBoost when its inputs are noisy predicted probabilities. Model architecture should therefore be matched to the representation and validated empirically.

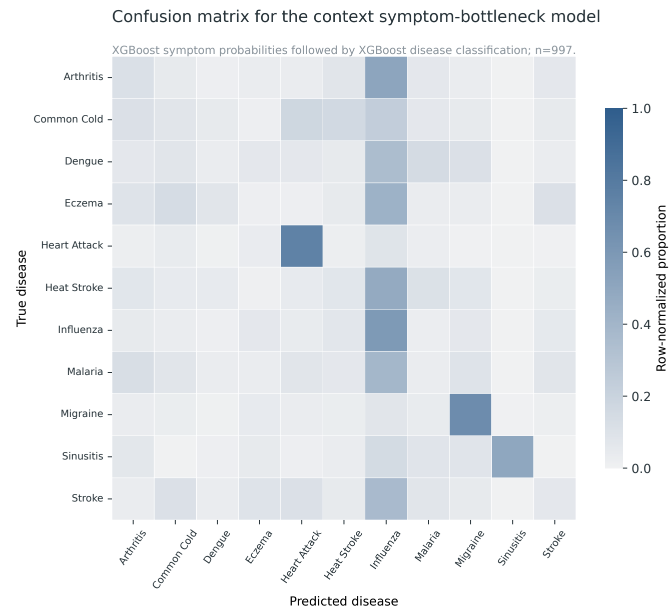

Fig. 5. Row-normalized locked-test confusion matrix for context-predicted XGBoost symptom probabilities followed by XGBoost disease classification.

C. Implications for additional experiments

The highest-value next experiment is not another random internal split. It is external validation on a temporally ordered cohort with patient, time, location, station, and institution identifiers. Before accessing external outcomes, the current feature construction, thresholds, and model configurations should be frozen. The external study should report cohort flow, missingness, class prevalence, distribution shift, subgroup performance, and calibration with confidence intervals. Any recalibration should be evaluated on a second untouched cohort.

VII. Limitations

First, the public file does not expose how participants, medical records, weather stations, symptoms, or disease outcomes were sampled and linked. Second, weather-signature grouping cannot replace patient-level or temporal separation. Third, duplicate removal may not eliminate non-exact repeated participants. Fourth, symptoms may be contemporaneous with or downstream of diagnosis, so the Oracle must not be interpreted as a prospective predictor. Fifth, the dataset is class-imbalanced, and several labels have only about 300 unique rows. Sixth, the humidity-unit decision is inferred from stored values rather than a formal data dictionary. Seventh, neither internal calibration nor feature attribution establishes clinical utility or causality. Finally, the context extension uses fields labeled as pre-existing conditions in the source, but their temporal availability cannot be verified.

VIII. Conclusion

A semantic weather–symptom–disease cascade is interpretable in structure but not automatically reliable in operation. Under a duplicate-removed, weather-group-disjoint,

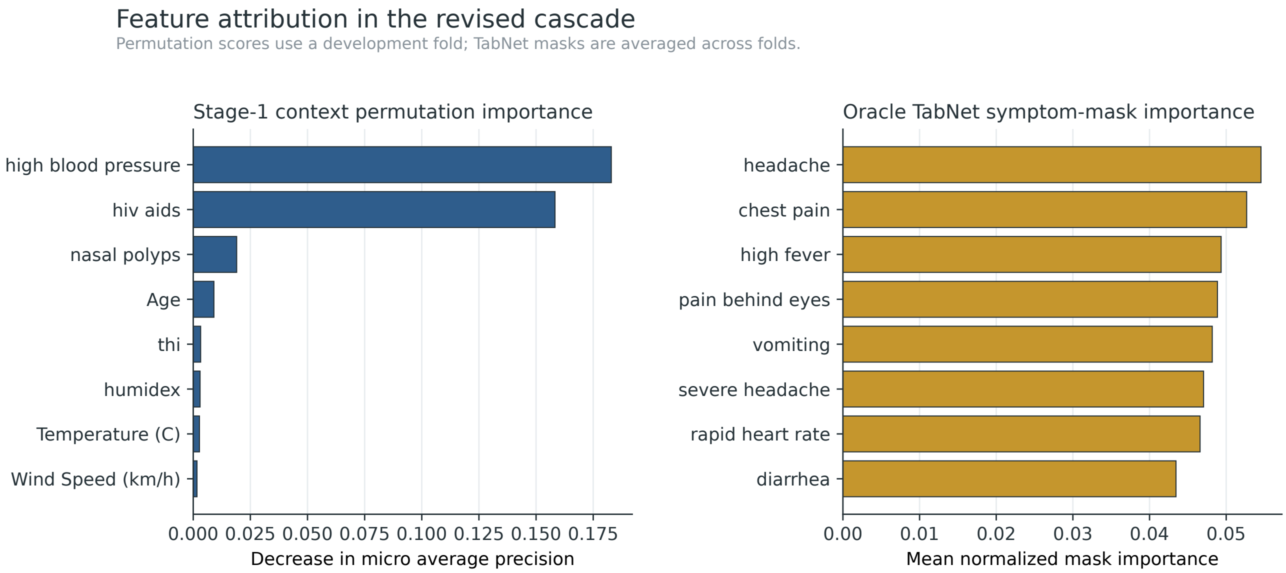

Fig. 6. Development-fold permutation importance for context models and aggregated Oracle TabNet masks. Importance measures are model- and representation-specific.

OOF-trained evaluation, weather-only symptom prediction is weak, and the deployable XGBoost-to-TabNet cascade falls below the class-prior baseline. True symptoms support high disease classification, localizing the failure to the intermediate prediction step. Pre-event context improves both direct and bottleneck models, but the direct context classifier remains strongest. These findings support the project as a reproducible model audit and error-propagation study. They do not support deployment, diagnosis, or causal claims without independently collected external validation.

Reproducibility and Data Availability

The raw dataset is available from Zenodo at https://doi.org/10.5281/zenodo.11366485. The accompanying project records the official checksum, configuration, package versions, split indices, out-of-fold and test probabilities, thresholds, metrics, confidence intervals, statistical comparisons, and figure-generation scripts. The publication protocol uses random seed 2026 and 2,000 bootstrap replicates with seed 2407.

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